

# 2110211 Introduction to Data Structure

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## Lecture 1

### C++ Intro

variable

### C++ from Java & Python

data

C++, if, for, while, function

scale

qwertyuio

## Lecture 2

### Array & Vector

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> a = {2,3};
    vector<bool> b = {true,false,true};

    a.push_back(10);
    a.push_back(20);

    a.insert( a.begin(), 99);
    a.insert( a.end(), -5);

    sort(a.begin(),a.end());

    cout << "a is ";
    for (size_t i = 0; i < a.size(); i++) {
        cout << a[i] << " ";
    }
    cout << endl;

    cout << "b is ";
    for (size_t i = 0; i < b.size(); i++) {
        cout << b[i] << " ";
    }
    cout << endl;
}
```

### Word Count Problem

```
#include <iostream>
#include <fstream>
#include <string>
#include <vector>
#include <set>
```

```
#include <algorithm>
#include "Tokenizer.h"

using namespace std;

void printUniqueWords1(string filename);
void printUniqueWords2(string filename);
void printUniqueWords3(string filename);
void printUniqueWords4(string filename);
void printWord(string filename);

int main() {
    string filename = "test.txt";
    printWord(filename);
    printUniqueWords1(filename);
    printUniqueWords2(filename);
    printUniqueWords3(filename);
    printUniqueWords4(filename);
    return 0;
}

bool search(string words[], int n, string w) {
    for (int i = 0; i < n; i++) {
        if (words[i] == w) return true;
        n++;
    }
}

void printWord(string filename) {
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        n++;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

void printUniqueWords1(string filename) { // wastes too much memory
    string words[100000];
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) words[n++] = token;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

## Dynamic Array

```
void printUniqueWords2(string filename) { // expandable array; wastes time
    int cap = 1;
    string *words;
    words = new string[cap];
    int n = 0;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) {
            if (n == cap) {
                string *a = new string[2*cap];
                for (int i=0; i<n; i++) a[i] = words[i];
                delete[] words;
                words = a;
                cap *= 2;
            }
            words[n++] = token;
        }
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

## Word Count by Vector & Set

### Vector

```
void printUniqueWords3(string filename) { // slow find time
    vector<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (words.end() == find(words.begin(), words.end(), token))
            words.push_back(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

### Set

```
void printUniqueWords3(string filename) { // quick find existence
    set<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        // if (words.end() == find(words.begin(), words.end(), token)) // slow find
        if (words.end() == word.find(token)) // if can't find then return word.end()
            words.insert(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}
```

## Lecture 3

### std::pair

#### Basic

```
#include <iostream>

int main() {
    std::pair<int, float> x;
    std::pair<int, float> y;

    x.first = 10;
    x.second = 20.65;

    std::cout << x.first << " "; // 10
    std::cout << x.second << std::endl; // 20.65

    y = x;

    std::cout << y.first << std::endl; // 10
    std::cout << y.second + 20 << std::cout; // 40.65
}
```

Pair reserves 2 spaces in RAM for first and second elements. When you use `y = x`, it copies values of `x` to `y`. You need to build template by specifying data types in a pair.

#### Initialize

```
#include <iostream>

int main() {
    // default constructor
    std::pair<std::string, bool> p; // default of that type: {"", false}
    std::cout << "default [" << p.first << "]" << p.second << "]" << std::endl;

    // initialize by { }
    std::pair<std::string, bool> p1 = { "somchai", true };

    // create pair without specifying type by "make_pair"
    std::pair<bool, int> p2;
    p2 = std::make_pair(false, 10);

    std::pair<bool, int> p3(p2); // must be same type p3 = p2

    // more complex pair
    std::pair< std::pair<float, int>, std::string > p4 = { {20.5, -3}, "abc"};
    std::cout << p4.first.first << " " << p4.first.second << " " << p4.second <<
std::endl; // 20.5 -3 abc
}
```

#### Array inside Pair

```
#include <iostream>

int main() {
    std::pair<std::string*, std::pair<int, float>> p;

    p.first = new std::string[5];
```



```

    p.first[0] = "a";
    p.first[1] = "b";
    p.first[2] = "c";

    p.second = std::make_pair(10,1.5);

    std::cout << p.first[0] << " " << p.second.first << " " << p.second.second;
}

```

## Template

- Template allows “same code, different data types”
- `std::pair` is a “class template”
  - In generic term, pair is defined as `pair<T1, T2>`
- To use `std::pair`, we must supply “Type information” to the template what T1 and T2 should be.

## STD and Namespace

- The “Fullname” of `cout` is `std::cout`
- `std` is a namespace
- We need to use fullname
  - Too lazy? use “using” keywords

```
#include <iostream>
```

```
// this tells C++ that when we say cout, we mean std::cout;
using std::cout;
```

```
int main() {
    // we still need to use std::endl
    cout << "Yes" << std::endl;
}
```

```
#include <iostream>
```

```
// this tells C++ that when we say something that it does not understand, C++ should
try to use std as its namespace
```

```
// this is VERY BAD PRACTICE in real world.
// 10/10 not recommend
// ... but it's ok for this class
using namespace std;
```

```
int main() {
    cout << "Yes" << endl;
}
```

## std::vector

A linear storage of a single data type

### Basic

```
#include <iostream>
#include <vector>
```

```
using namespace std;
```

```
int main() {
```

```

vector<int> v1;
cout << "Size of v1 is " << v1.size() << endl; // 0

vector<int> v2 = {2,3,4};

cout << v2[1] << endl; // 3
v1 = v2;
v1[0] = 20;

cout << v1[0] << ", " << v1[1] << v1[2] << endl; // 20, 3, 4

v1.push_back(99);
cout << v1.size() << endl;
}

```

- Vector starts with empty element
- Use size() to get the number of element
- Use empty() to check if a vector has any element

### Initialize

- With specific size
- With specific size and starting value

```

#include <iostream>
#include <vector>

using namespace std;

void print_vector(vector<float> v) {
    for (size_t i = 0; i < v.size(); i++) {
        cout << v[i] << " ";
    }
    cout << endl;
}

int main() {
    vector<float> v1(10); // size = 10
    print_vector(v1);

    vector<float> v2(5, 3.55); // size = 5, default = 3.55
    print_vector(v2);

    vector<float> v3(v2);
    print_vector(v3);
}

```

### Vector Access & Overflow Problem

#### Access

```

#include <iostream>
#include <vector>

using namespace std;

int main() {
    vector<float> v1(2);
    vector<float> v2(2);
}

```

```

cout << "-- v1 --" << endl;
for (int i = 0; i < 7; i++) {
    v1[i] = i; // this automatically assigns value
    cout << i << ": " << v1[i] << endl; // print nonsense if out of range
}
cout << "-- v2 --" << endl;
for (int i = 0; i < 7; i++) {
    cout << i << ": " << v2[i] << endl; // print "4 5 6" due to buffer overflow (take
space of v1)
}
cout << "using at" << endl;
v2.at(1) = 99;
// this will cause exception 'std::out_of_range'
for (int i = 0; i < 7; i++) {
    cout << ": " << v2.at(i) << endl;
}
}

```

- Operator [] won't check for 'out-of-range'
  - Reading, writing beyond size is undefined
  - Might crash
  - Grader will give 'x'
- at () will check bound
  - But slower

You should be careful of using spaces, it can access illegal memories and cause weird behaviors.

## Modifying Vector

### Resizing

- Resize change the size
  - Enlarge will fill with default

```

#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ":";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,10);
    print(v); // 10,10,10,
    v.resize(6);
    print(v); // 10,10,10,0,0,0,
    v.resize(1);
    print(v); // 10,
    v.resize(7);
    print(v); // 10,0,0,0,0,0,0
}

```

**Modify**

- pop\_back
  - Erase last element
- insert(position,value)
- erase(position)
- Careful!
  - Both insert and erase position must be valid
  - If not valid, it can crash

```
#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,8); // v.begin() is called iterator
    v.insert( v.begin(), 1); // 1, 8, 8, 8
    v.insert( v.begin()+3, 2); // 1, 8, 8, 2, 8
    v.insert( v.end(), 3); // 1, 8, 8, 2, 8, 3
    print(v);
    v.erase(v.begin()); // 8, 8, 2, 8, 3
    v.erase(v.begin()+2); // 8, 8, 8, 3
    print(v);
    v.pop_back(); // 8, 8, 8
    print(v);
}
```

**std::find & std::sort****Find**

- find(a,b,c)
  - a and b are position (iterator)
  - find c from a to *before* b
  - if not found return b
- Must #include <algorithm>

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    int x = 5;
    if ( find(v.begin(), v.end(), x) != v.end() ) {
        cout << "found" << endl;
    } else {
        cout << "not found" << endl;
    }
}
```

```

}

if (find(v.begin(), v.begin()+3, 3) != v.begin()+3) cout << "FOUND" << endl;

// How many "YES" will be printed? (CHEAT QUESTION)
if (find(v.begin(), v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin(), v.begin()+4, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+8, x) != v.end()) cout << "YES" << endl; // YES
// First 3 can't even give v.end(), last did find so it gave that position instead
}

```

### Sort

```

#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    print(v); // 9, -1, 3, 7, 5, 2, 1, 4,
    sort(v.begin()+2, v.begin()+6);
    print(v); // 9, -1, 2, 3, 5, 7, 1, 4,
}

```

### Functions that Work with Vector

- sort
- find
- min\_element
- max\_element
- lower\_bound
- upper\_bound

Need to use *iterator*

## Lecture 4

### Vector Iterator

#### Iterator

- Iterator is a *pointer to position*
- First element is `begin()`
- The one *after* the last element is `end()`
- For insert, valid position is from `begin()` to `end()`, inclusive
- For erase, valid position is from `begin()` up to *before* `end()`
- Different vector, different iterator
  - `v1.end()` is not the same as `v2.end()`

### Iterator Arithmetic

- We can add by integer to an iterator
- We can subtract two iterators of the same type
- We can use increment (++) and decrement (--)

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    vector<int>::iterator it1 = v1.begin() + 3; // 30
    vector<float>::iterator it2 = v2.end();      // after 2.73

    // getting value at iterator by using "*" operator
    cout << *it1 << endl;      // 30
    cout << *(it2-1) << endl; // 2.73
    cout << *it1+2 << endl;    // 32

    // iterator arithmetics
    vector<int>::iterator it3 = it1 + 2;
    cout << "data at it3: " << *it3 << endl; // 50
    cout << "difference of it3,it1: " << (it3 - it1) << endl; // 2

    vector<float>::iterator it4 = v2.end();
    it4--;
    cout << "data of it4: " << *it4 << endl; // 2.73

    // this cannot be done
    // cout << (it2 - it1) << endl;
}
```

### Iterator All Elements by Iterator

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    cout << "----v1----" << endl;
    auto it = v1.begin();
    while (it < v1.end()) { // should use != in other iterator types
        cout << it - v1.begin() << ": " << *it << endl;
        it++;
    }

    cout << "----v2----" << endl;
    for (auto it = v2.begin(); it < v2.end(); it++) {
        cout << it - v2.begin() << ": " << *it << endl;
    }
}
```

- We can compare iterator
  - Beware!! Iterator of some data structure does not support > or < comparator
  - But for vectors, it's ok
- We can use auto keyword to automatically define a type of a variable

## Iterator Invalidate

### Some Functions that Use Iterators

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
    vector<int> v1 = {3,0,1,2,4,-3,9,8};
    vector<float> v2 = {10.2, -4, 0.13, 3.14, 2.73};

    auto it1 = min_element(v1.begin(),v1.end());
    auto it2 = max_element(v2.begin()+2,v2.end());

    cout << *it1 << endl; // -3
    cout << *it2 << endl; // 3.14
}
```

- min\_element and max\_element get the iterator of min, max element

### New Idiom that Iterate All Elements

- Shorter syntax for loop
- Can use reference operator (&)

```
#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
using namespace std;

int main() {
    vector<string> v1 = {"somchai", "somying", "somsak"};

    // range-based for loop
    for (string x : v1) {
        // x is a copy of each element in v1
        cout << x << ", ";
    }

    // using reference
    // x is THE element of v1, meaning we can modify it
    for (auto &x : v1) { x.replace(0,4,"--"); }
    for (auto &x : v1) { cout << x << " "; } // --hai --ing --ak
    cout << endl;
}
```

### Iterator Invalidation

```
#include <iostream>
#include <vector>
using namespace std;
```

```
int main() {
    vector<int> v1 = {10,20};

    auto = v1.end()-1; // this points to 20
    // we resize (enlarge) v1, now [it] is invalidated
    v1.resize(10);

    // this might not crash but it actually points to somewhere not in v1
    cout << *it << endl;

    // this will crash the program
    // v1.insert(it,99);
    for (auto &x: v1) {cout << x << " ";}
}
```

- Some operation on the data structure *invalidate* existing iterators
  - For vector, this includes *addition* and *deletion* of element (including *resize*)

## Set

- Storing distinct data of same type
  - The data type must be *comparable*, i.e., we can tell if a is more or less than b
- Somewhat slow insert
- Fast look up
- Fast erase
- Iterator starts from “minimum element” and goes in increasing value direction
  - Can be used to (somewhat) fast storing

## Basic

- Notice that s does not include duplicate elements
- Also see that when we iterate, member is sorted
  - This is distinct characteristic of set

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set<int> s = {4,1,3,2,1,1,3,4};

    cout << "Size of s is " << s.size() << endl; // 4

    s.insert(10);
    s.insert(5);
    s.erase(3);

    cout << "member of s: "; // 1 2 4 5 10
    for (auto it = s.begin(); it != s.end(); it++) {
        cout << *it << " ";
    }
}
```

## Somewhat Slow Insert, Iterate but Fast Find

- We will see this in the detail around last part of this course
  - For now, please believe that
    - If there are  $N$  elements in the set



- Insert takes time directly proportional to  $\log(N)$
- Each `it++` or `it--` takes times directly proportional to  $\log(N)$
- Each find takes time directly proportional to  $\log(N)$

## Set Iterator

### Demos Comparing Vector & Set

- Insert
  - Vector is faster than set
- Find random number
  - Set is much faster than vector

### Set Iterator

- We cannot do `s.begin() + x`
  - Because, going to the next element (which is the *successor*) in set is not as fast as vector, c++ forbids `begin() + x`
    - alternatively, you can use `auto it = s.begin(); for (int i = 0; i < 0; i++) {it++;}`
  - We cannot *compare by* `>` or `<`
- We can still use `it++` or `it--` to go to the next or previous (*successor* or *predecessor*) or `x`

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set< pair<string,int> > s = { {"somchai",5}, {"abc",6}, {"abcd",-3}, {"somchai",-4},
                                {"z",0}, {"z",-1}, {"z",9} };

    for (auto &x : s) {
        cout << x.first << "," << x.second << endl;
    }

    cout << "-- find --" << endl;
    auto it = s.find({"z",-1});
    cout << (*it).first << "," << (*it).second << endl;
    it--;
    it--;
    cout << it->first << "," << it->second << endl;
    it++;
    cout << it->first << "," << it->second << endl;
}
```

### Additional Function

- `set.lower_bound`
- `set.upper_bound`
- `set.count`

## std::map

Associative data structure with same property as set

### Map

- Is very similar to Python's dict in usage
- Is internally implemented as a set with "pair" data type
  - *Same properties, same limitations* as set but more convenience to use as associative data structure
- Associative (mapping) between a *Key Type* and a *Mapped Type*

**Basic**

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    // map between "Key Type" string and "Mapped Type" int
    map<string,int> m;
    m["somchai"] = 10;
    m["somying"] = -5;
    cout << "Size = " << m.size() << endl; // 2

    // accessing unseen key create a map with default value
    cout << "m[\"xxx\"] = " << m["xxx"] << endl; // 0
    // each element is a pair of key type and mapped type
    for (auto it = m.begin(); it != m.end(); it++) {
        cout << it->first << " is mapped to " << it->second << endl;
    } // somchai: 10, somying: -5

    // this will create mapping "abc" to 0 first and then increase it
    m["abc"]++;
    cout << "now size = " << m.size() << endl; // 4
    for (auto &x : m) {
        cout << x.first << " is mapped to " << x.second << endl;
    } // abc: 1, somchai: 10, somying: -5, xxx: 0
}
```

**Checking If Map Has This Key?**

- Use find()

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    map<int,string> m;
    m[1] = "somchai";
    m[99] = "nattee";

    int k = 99;
    map<int,string>::iterator it;
    if ((it = m.find()) != m.end()) {
        cout << "Key " << it->first << " is mapped to " << it->second << endl;
    } else {
        cout << "Key " << k << " does not exist in m." << endl;
    }

    // this is not the correct way to check if key exists why??
    if (m[k] != "") {
        cout << "exists" << endl;
    } else {
        cout << "does not exists" << endl;
    }
}
```

**Requirement of std::set and std::map**

- Set data type and map key type must be comparable
  - We must be able to compare *order* of two elements
- Type that we can use directly
  - int, bool, float, string, double, char, ... and most of other numerical data type
  - Pair can also be used if both the types of first and second are comparable
    - Pair compare first then second

**Practice Reading C++ Docs**

- Both map and set has *insert* and *erase* function
- What is the return value of both function of each data structure?
  - For `set<int> s`, can we do `s.erase(20)`
  - For `map<string, bool> m`, can we do `m.erase("Somchai")`??
- If we wish to erase element from index 3 to index 4096 in a vector
  - Is there any function from vector that we can easily use?

**Pair-Sum Problem**

- Pair Sum
  - Given an array of integers, our task is to find whether there exists a pair of elements in the array such that their summation equal to  $X$
- Input:
  - Array of integers (our main array); this is a large array
  - $M$  values of  $X$ , for each value  $X$ , we have to determine if a pair whose sum equal to  $X$  exists.
- Output:
  - For each value of  $X$ , print "YES" if we found such pair; print "NO" otherwise

**Answer**

```
#include <iostream>
#include <array>
#include <algorithm>
#include <cmath>

using namespace std;

bool check(int a[], int n, int X);

int main() {
    int a[15] = {323, 232, 138, 230, 437, 176, 127, 111, 329, 1, 296, 12, 3, 2, 6};
    int n = sizeof(a) / sizeof(a[0]);
    int X = 10000;

    cout << check(a,n,X);
}

bool check(int a[], int n, int X) {
    for (int i=0; i<(pow(2,n)); i++) {
        int t = X;
        for (int j=0; j<n; j++) {
            if (i / ((int) pow(2,j+1)) % 2 == 0) {
                t -= a[j];
            }

            if (0 == t) {
                cout << "YES" << endl;
            }
        }
    }
}
```

```

        return true;
    }
    if (0 > t) {break;}
}
}
cout << "NO" << endl;
return false;
} // My stupid solution where I brute force every possible combinations

```

## Lecture 5

### Stack

การเพิ่ม / ลบข้อมูลในกองซ้อน

- ข้อมูล เข้าหลัง ออกก่อน (Last\_In First-Out)

กองซ้อน : **stack**

```

bool        empty();
unsigned    size();
T           top();           // ดูปบนสุด
void        push(T element); // ใส่ข้อมูล
void        pop();           // เอาออก

```

ข้อควรระวัง

- ใช้ `#include <stack>`
- `top()` ตอนไม่มีข้อมูล → พัง
- `pop()` ตอนไม่มีข้อมูล → พัง
- stack ไม่มี iterator
  - ดูได้เฉพาะข้อมูลที่เพิ่งใส่เข้าไปล่าสุด (`top`)
  - ไม่มี `begin()`, `end()` ให้ใช้
  - ถ้าอยากมาดูข้อมูลทุกตัวใน stack ต้องจำใจ `pop` ข้อมูลออกมา

ตัวอย่างการใช้งาน **Stack**

- การตรวจสอบโครงสร้างแบบซ้อนกัน เช่น การใส่วงเล็บ `(){}[]...`
- การจัดเก็บตัวแปรและการทำ `function calls`
- การประมวลผลนิพจน์ทางคณิตศาสตร์ `postFix` → Discrete Math
- การทำ `undo/redo`
- การค้นคำตอบแบบ `depth-first search`
- ...

### Parentheses Checking

การตรวจสอบการใส่วงเล็บ

- ถูก : `( { ( ) [ { } ] } )`
- ผิด : เปิดปิดไม่ตรงกัน `( { ] )`
- ผิด : มีปิดมากเกินไป `( { ( ) } ) ) }`
- ผิด : มีเปิดมากเกินไป `( { ( ) }`
- วิธีทำ
  - อ่านมาทีละตัว
  - ถ้าเป็นวงเล็บเปิด ให้ `push` ลง stack
  - ถ้าเป็นวงเล็บปิด ให้ `pop` จาก stack มาตรวจสอบว่าเป็นวงเล็บเปิดที่ตรงกันกับวงเล็บปิดหรือไม่
  - เมื่อใดที่อยาก `pop` ถ้า `isEmpty` แสดงว่า ปิดมีมากเกินไป

- ▶ เมื่ออ่านเสร็จหมด stack ยังมีข้อมูล แสดงว่า เปิดมีมากไป

## Function Call Stack

### Using Stack in Java Virtual Machine

- jvm uses java stack to store
  - ▶ state of method calls
  - ▶ parameters and local variables
  - ▶ temporary memory for computation (operand stack)

```
public class StackFrame {
    public static void main(String[] args) {
        a(3,2);
        b(5);
    }
    static void a(int x, int y) {
        int z = x/y;
        b(z);
    }
    static void b(int x) {
        ++x;
    }
}
```

We make a stack frame of function calls, each element consists of arguments and returning argument of the function. Function main contains args and RA JVM and calls a, a will be added to stack and contains z, y, x, and RA 04:, then it calls b. After b is finished, it will come back to line 9 and popped itself. It will do a and go to line 4, and so on. Arguments in each functions are separated variables.

```
#include <iostream>

using namespace std;

void test(int x) {
    if (x > 0) {
        int y = x-1;
        cout << "I am test(" << x << "). continue... " << endl;
        test(y);
        cout << "now x is " << x << endl;
    } else {
        cout << "I am test(0). stop"
    }
}

int main() {
    test(4);
    return 0;
}
```

### Output

```
I am test(4). continue...
I am test(3). continue...
I am test(2). continue...
I am test(1). continue...
I am test(0). stop
now x is 1
now x is 2
```

now x is 3  
now x is 4

## Postfix Evaluation

### Infix and Postfix Expressions

- infix (เดิมกลาง)
  - $a + b * c / d - 2, (a + b) * c / (d - 2)$
  - use order of operations
  - use parentheses
- postfix (เดิมหลัง)
  - $a b c * d / + 2 -, a b + c * d 2 - /$
  - order is from left to right
  - parentheses may be omitted

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | + | 3 | * | 4 | 1 | - | / |   |
|   |   |   | 3 | 3 | * | 4 | 1 | - | / |
|   |   |   |   | 9 | 4 | 1 | - | / |   |
|   |   |   |   | 9 |   |   | 3 | / |   |
|   |   |   |   |   |   |   |   |   | 3 |

### Calculate Postfix Expressions with Stack

- Value of  $2 3 + 4 5 - 6 * + ?$
- Use stack to help
- Solution
  - Check each element in postfix from left to right
  - If it is operand, push stack
  - If it is operator, pop elements from stack as many as operator needs, then push the result
  - After complete, answer will be at the top of stack

## Convert Infix to Postfix

### Use Stack to Convert

- input : infix expression
- output : postfix expression
- Steps
  - Read each element in infix
  - If it is operand, move it to the back of output
  - If it is operator
    - May pop operator to the end of output
    - push operator to stack
  - If all infix is read
    - pop all operators to the end of input

```
string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) { // Read each element
        char token = infix[i];
        if (token เป็น operand) {
            postfix += token; // if operand, add to output
        } else {
            while ( ( ??? ) ) {
                postfix += s.top(); // if operator, pop stack to result,
                s.pop();
            }
        }
    }
    while (!s.empty()) {
        postfix += s.top();
        s.pop();
    }
    return postfix;
}
```

```

    }
    s.push(token);          // then push new operator
}
}
while (!s.empty()) {postfix += s.top(); s.pop();}
return postfix;
}

```

### Priority of Operators

input    2    \*    3    +    4

| read | output    | stack                                           |
|------|-----------|-------------------------------------------------|
| 2    |           |                                                 |
| *    | 2         |                                                 |
| 3    | 2         | *                                               |
| +    | 2 3       | * (* comes before and is more important than +) |
| 4    | 2 3 *     | +                                               |
|      | 2 3 * 4   | +                                               |
|      | 2 3 * 4 + |                                                 |

input    2    +    3    \*    4

| read | output  | stack |
|------|---------|-------|
| 2    |         |       |
| +    | 2       |       |
| 3    | 2       | +     |
| *    | 2 3     | +     |
|      | 2 3     | * +   |
|      | 2 3 *   | +     |
|      | 2 3 * + |       |

### Code to Compare Priority

```

string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) {
        char token = infix[i];
        if (priority.find(token) != priority.end()) {
            postfix += token;
        } else {
            int p = priority[token]; // Get priority of the new operator
            while ( !s.empty() && priority[s.top()] >= p ) { // lazy evaluation; left first
                postfix += s.top(); // pop if top operator is not less important than new op
                s.pop();
            }
            s.push(token);
        }
    }
    while (!s.empty()) {postfix += s.top(); s.pop();}
    return postfix;
}

```

## Lazy Evaluation

### Setting Priority of Operators

- $\wedge$  represents power 7
- $\wedge$  before  $+$   $-$   $*$   $/$
- $*$   $/$  before  $+$   $-$
- $*$  same as  $/$
- $+$  same as  $-$

```
map<char,int> priority =
    {{'+',3},{'-',3},{'*',5},{ '/',5},{'^',7}};
```

### Complications with Parentheses

- In parentheses is like sub-expression
- Open parentheses always push (very high priority)
- Close parentheses is very low priority
- If found, pop until sees open parentheses

input    2    \*    (    3    +    4    )

| read | output    | stack           |
|------|-----------|-----------------|
| 2    |           |                 |
| *    | 2         |                 |
| (    | 2         | *               |
| 3    | 2         | ( *             |
| +    | 2 3       | ( *             |
| 4    | 2 3       | + ( *           |
| )    | 2 3 4     | + ( * (pop all) |
|      | 2 3 4 + * |                 |

```
int p = priority[token];
while ( !s.empty() && priority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
s.push(token);

int p = outpriority[token];
while ( !s.empty() && inpriority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
if (token == ')') s.pop(); else s.push(token);
```

### Priority of Operators with Parentheses

|               | + | - | * | / | $\wedge$ | ( | ) |
|---------------|---|---|---|---|----------|---|---|
| In expression | 3 | 3 | 5 | 5 | 7        | 9 | 1 |
| In stack      | 3 | 3 | 5 | 5 | 7        | 0 |   |

### Right to Left Operator

|               | + | - | * | / | $\wedge$ | ( | ) |
|---------------|---|---|---|---|----------|---|---|
| In expression | 2 | 2 | 4 | 4 | 8        | 9 | 1 |
| In stack      | 3 | 3 | 5 | 5 | 7        | 0 |   |



**Note** Power in expression is more important than in stack

input    2    +    3    -    4    ^    5    ^    6

| read | output            | stack |
|------|-------------------|-------|
| 2    |                   |       |
| +    | 2                 |       |
| 3    | 2                 | +     |
| -    | 2 3               | +     |
| 4    | 2 3 +             | -     |
| ^    | 2 3 + 4           | ^ -   |
| 5    | 2 3 + 4           | ^ -   |
| ^    | 2 3 + 4 5         | ^ -   |
| 6    | 2 3 + 4 5         | ^ ^ - |
|      | 2 3 + 4 5 6       | ^ ^ - |
|      | 2 3 + 4 5 6 ^ ^ - |       |

### Conclusion

- Implementations of stack to solve problems
- Main operations: push / pop / top
- Make stack with array
- If reserve enough space, all runs is  $\Theta(1)$ ; constant time

## Lecture 6

### Queue

- Just like a normal queue in real life.
  - First-in-First-Out data structure
  - One way in (back of queue), one way out (front of queue)
  - push = add data to the back of the queue
  - pop = remove data from the head of the queue

### Basic

```
#include <iostream>
#include <queue>
#include <vector>

using namespace std;

int main() {
    queue<int> q;
    q.push(1);
    q.push(2);
    q.push(3);
    while (q.empty() == false) {
        cout << q.front() << endl;
        q.pop();
    }
    cout << "-- example 2 --" << endl;
    queue<vector<int>> q2;
    vector<int> v1 = {1,2,3};
    vector<int> v2 = {99,88,-1};
    q2.push( v1 );
```

```

q2.push( v2 );
cout << q2.back()[1] << endl;
cout << q2.front().size() << endl;
auto x = q2.front();
q2.pop();
cout << x[0] << endl;
}

size_t    q.size()
bool      q.empty()
void      q.push(T data)
void      q.pop()
T         q.front()
T         q.back()

```

### Limitation

- Same limitation as stack
  - No iterator
    - No `begin()`, `end()`
    - Can only access front and back of the queue
    - If we wish to access all members, we have to pop it all
  - Do not call `front()`, `back()`, `pop()` when the queue is empty

## Radix Sort

Queue Application: Fast sorting with no comparison

### Overview

- Put all data in an array
- For each digit  $X$ , from LSD to MSD
  - PUT TO QUEUE step: Sort by digit  $X$  by putting all of data from the array into  $B$  queues
    - $B$  is the base of the number
    - For example, for a base 10 number, we will have Queue[0] to Queue[9]
    - Put into the queue labelled with that digit
  - GET FROM QUEUE step: Start from queue 0 to queue  $B - 1$ , remove data from the queue and put back to the array

### Example

Input 115 15 42 305 21 8 463

#### Round 1 (digit 0), from queue

| (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|     |     |     |     |     | 305 |     |     |     |     |
|     |     |     |     |     | 15  |     |     |     |     |
|     | 21  | 42  | 463 |     | 115 |     |     | 8   |     |

⇒ 21 42 463 115 15 305 8

Data is now sorted by the last digits, because we pop from queue 0 to 9

#### Round 2 (digit 1), from queue

| (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 8   | 15  |     |     |     |     |     |     |     |     |
| 305 | 115 | 21  |     | 42  |     | 463 |     |     |     |

⇒ 305 8 115 15 21 42 463

Data is now sorted by last two digits, because we goes from queue 0 to 9, which is grouped by digit 1 and the data in each queue is sorted by the last digit.

### Round 3 (digit 2), from queue

| (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 42  |     |     |     |     |     |     |     |     |     |
| 21  |     |     |     |     |     |     |     |     |     |
| 15  |     |     |     |     |     |     |     |     |     |
| 8   | 115 |     | 305 | 463 |     |     |     |     |     |

⇒ 8 15 21 42 115 305 463

### Code

```
#define base 10

int getDigit(int v, int k) {
    // return the kth digit of v (MSD is digit 0)
    int i;
    for (int i=0; i<k; i++) v /= base;
    return v % base;
}

// d = number of digits
void radixSort(vector<int> &data, int d) {
    queue<int> q[base];
    for (int k=0; k<d; k++) {
        for (auto &x : data) {
            q[getDigit(x,k)].push(x);
            for (int i=0, j=0; i<base; i++) {
                while(!q[i].empty()) {
                    data[j++] = q[i].front(); q[i].pop();
                }
            }
        }
    }
}
```

## Breadth First Search

Queue Application: Gotta generate 'em all

### The Problem

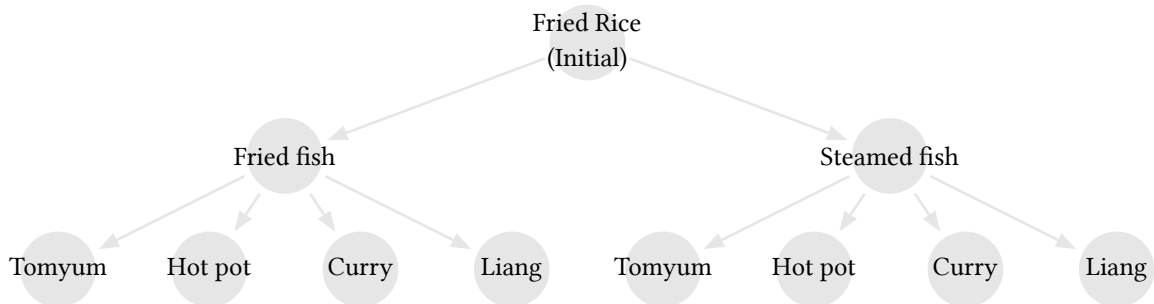
- Given a positive integer  $X$
- Start with a number 1, find a sequence of arithmetics operations, either “ $\times$  3”, or “ $/$  2” that makes 1 into  $X$ 
  - the  $/$  2 is integer division, e.g.,  $5 / 3 = 1$  (not 1.6666)
  - The sequence must be as short as possible
- Example
  - $10 = 1 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2$
  - $31 = 1 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2 / 2 / 2 \times 3 \times 3 / 2$

### The Idea

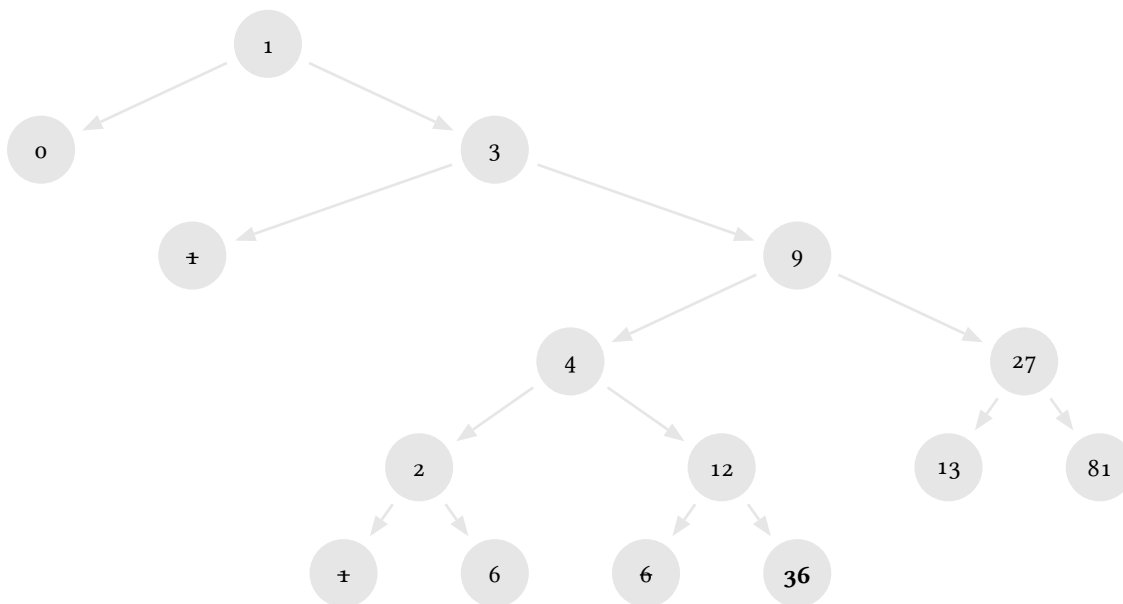
- Generate all possible sequences
  - Start with length 1, 2, 3, ... until we find one that gives  $X$
- This is called an *exhaustive search* algorithm
  - Systematically enumerate all possible somethings

**Tree Structure (Search Tree)**

- A structure to illustrate *search* algorithm
- Divide into steps
  - Start with *initial solution*
  - For each possible outcome (called a *state*) of each step, *generate all* proper possible *next step*
    - Also, check if the current step is what we needed
- Written as a diagram of *node* and *edge*

**Enumerate****M3D2 Problem & Solution****Back to Our Problem**

- Start with 1
- Each step is either \* 3 or / 2
- Issue: might get repeated number
  - Solution: if we have found it, do not generate new step

**M3D2 Code****Code**

- `Queue` makes ordering of how we pick a state to enumerate
- From top to bottom and left to right

```

void m3d2(int target) {
    map<int, int> prev;
    queue<int> q;
  
```

```

int v = 1;
q.push(1); prev[1] = -1;
while (!q.empty()) {
    v = q.front(); q.pop();
    if (v == target) break;
    int v2 = v/2;
    int v3 = v*3;
    if (prev[v2] == 0) {q.push(v2); prev[v2] = v;}
    if (prev[v3] == 0) {q.push(v3); prev[v3] = v;}
}
if (v == target) showSolution(v, prev);
}

```

### Display Solution

|         |   |    |   |   |   |   |   |    |    |    |    |    |    |
|---------|---|----|---|---|---|---|---|----|----|----|----|----|----|
| x       | 0 | 1  | 2 | 3 | 4 | 6 | 9 | 12 | 13 | 27 | 36 | 39 | 40 |
| prev[x] | 1 | -1 | 4 | 1 | 9 | 2 | 3 | 4  | 27 | 9  | 12 | 13 | 81 |

### showSolution

```

void showSolution(int v, map<int,int>& prev) {
    string out = "";
    while (prev[v] != -1) {
        if (prev[v] * 3 == v) {
            out = "x3" + out;
        } else {
            out = "/" + out;
        }
        v = prev[v];
    }
    out = "1" + out;
    cout << out << endl;
}

```

⇒ 36 = 1 \* 3 \* 3 / 2 \* 3 \* 3

## Lecture 7

### Priority Queue

Queue with privilege

#### Intro

- Priority Queue is ...
  - A queue with priority
  - Item with high priority is promoted to the front of the queue
    - There is *no back of the queue*
  - Priority is defined by having more value
    - Comparison, by default, is to use operator <, i.e., if item A < B is true, then B *has higher priority*
    - We can have custom comparator
- Has the *same interface* as stack

#### Example

For intuitive purpose only! This is not really how priority\_queue work internally

```

push(10)
< 10      >

```

```
push(30)
< 30 10 >
```

```
push(20)
< 30 20 10 >
```

```
pop()
< 20 10 >
```

```
push(15)
< 20 15 10 >
|
top
```

### Basic

```
size_t      q.size()
bool        q.empty()
void        q.push()
void        q.pop()
T           q.top()
* there is no q.back()
```

### Limitation

- Same limitation as stack, queue
  - *no iterator*
    - No `begin()`, `end()`
    - Use `#include <queue>`
    - Can only access `top()` of the queue
    - If we wish to access all members, we have to pop it all
  - Do not call `top()`, `pop()` when the queue is empty
- The data type must be *comparable* (similar to set and map)

## Class

### Quick Summary

- Syntax
  - class declaration must end with ;
  - Function definition can be outside of the class
  - Access modifier is `public:`, `private:`, `protected:`
  - Constructor is a function with the same name of the class with no return type
- Object is a variable (instantiation) of a class
  - When declared, a constructor is called

### Example 1

```
#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {
```

```

        return "[" + name + " " + surname + "];";
    }
private:
    string name, surname;
};

int main() {
    Student a;
    Student b;
    a.setFullName("nattee", "niparnan");
    cout << a.getFullName() << endl;
    cout << b.getFullName() << endl;
}

```

Declaring class before writing functions

```

class Student {
public:
    void setFullName(string name, string surname);
    string getFullName();
private:
    string name,surname;
};

void Student::setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
}

string Student::getFullName() {
    return "[" + name + " " + surname + "];";
}

```

### Output

```

[nattee niparnan]
[ ]

```

### Example 2: Constructor

```

#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    Student(float score) {gpax = score;}
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {return "[" + name + " " + surname + "];"}
    bool is1stHonor() {return gpax >= 3.6;}
private:
    string name, surname;
    float gpax;
};

int main() {

```

```

Student a(2.95);
a.setFullName("nattee", "niparnan");

cout << a.getFullName() << endl;
if (a.is1stHonor()) {cout << "YES" endl;} else {cout << "NO" << endl;}
// Student b; // <-- cannot compile because there is no default constructor
}

```

## Operator Overloading

How C++ has a function for each operator

### Overview

- Let say we write `a + b` when `a` and `b` is an object of some classes
  - This can be considered the same as calling a function `plus(a,b)`
  - C++ allows us to write a function for many operator and use it as an operator
  - For example we can write a function `times(a,b)` and let it be used as `a * b`
- This is what we called operator overloading

Instead of writing

```

Matrix multiply(Matrix a1, Matrix a2) {
    multiply(a, multiply(b,c))
}

```

We want to write

```

Matrix a, b, c
a * (b * c)

```

### Example

```

#include <queue>
#include <iostream>
#include <string>

using namespace std;

// use operator + symbol that will be overloaded
string operator*(string & lhs, const int & rhs) {
    string result = "";
    for (int i=0; i<rhs; i++) {
        result = result + lhs;
    }
    return result;
}

int main() {
    string a = "abc ";
    cout << (a * 3) << endl; // this gives "abc abc abc "
}

```

- Function must be named operator followed by the operator that we will overload
- Some operator takes two parameters (such as `+`, `-`, `*`, `/`, `%`)
- Some takes one (such as `++`, `--`, `!`, `*`, `&`)



## Overloading less-than operator

### Using with data structure that require sorting

- We have seen several data structure that requires comparability of the data, such as set, map and priority queue
- If we want to use our class with these data structure, we need to tell them how can we use them
- There are multiple ways to achieve this
  - Let us consider operator overloading

### Overloading <

- As stated earlier, set, map, and priority\_queue use *operator <* to compare two elements
- It does not work if we overload *operator >*

```
class Student {
public:
    Student(float score, string a, string b) {
        name = a;
        surname = b;
        gpax = score;
    }
    bool is1stHonor() {return gpax >= 3.6;}
    // not good, now our data is public
    string name, surname;
    float gpax;
    // overloading <
    bool operator<(const Student & other) const {
        return gpax < other.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    cout << (a < b) << endl; // 1
    priority_queue<Student> pq;
    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // attawith
}
```

## Custom Comparator

### Why custom?

- By overloading *operator <*, we have defined default ordering of that class
- What if we need another ordering, just for this *priority\_queue* only
  - For example, *Student* is ordered by *gpax* by default
  - What if we want our *priority\_queue* to order by name instead, while keeping the *Student* default ordering elsewhere
  - Better, can we have multiple *priority\_queue* with different ordering?
- Can be done via comparator class

### Example

```
#include <iostream>
#include <string>
#include <queue>
```

```

using namespace std;

class Student {...};

// Comparator class use () overloading
class StudentByNameComparator {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        return lhs.name < rhs.name;
    }
};

class GpaxThenName {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        if (lhs.gpax == rhs.gpax)
            return lhs.name < rhs.name;
        return lhs.gpax < rhs.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    Student c(4.00, "vishnu", "kotrajaras");
    cout << (a < b) << endl; // 1
    StudentByNameComparator comp1;
    GpaxThenName comp2;

    cout << comp1(a,b) << endl; // 0; can use like function

    // Use 3 template parameters
    priority_queue<Student, // class to compare
                  vector<Student>, // vector of that class
                  StudentByNameComparator> pq(comp1); // comparator class

    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // nattee

    priority_queue<Student,
                  vector<Student>,
                  GpaxThenName> pq2(comp2);

    pq2.push(a);
    pq2.push(b);
    pq2.push(c);
    cout << pq2.top().name << endl; // vishnu
}

```

#### Another Method, lambda-function

```

#include <iostream>
#include <string>
#include <queue>
using namespace std;

```

```
int main() {
    auto compare = [](const string& lhs, const string& rhs) {
        return lhs.size() < rhs.size();
    };

    cout << "Result of compare function = " << compare("xxx","z") << endl;

    priority_queue<string,vector<string>,decltype(compare)> pq(compare);
    pq.push("somchai");
    pq.push("z");
    pq.push("abc");
    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

### Output

```
Result of compare function = 0
somchai
abc
z
```

## Priority Queue Template

### Templating of priority\_queue

- priority\_queue requires 3 template parameters
- priority\_queue<T, Container = vector<T>, Compare = less<T>>
- The first one is required (which is the type of the data)
- The *second* and the *third* is optional (it has default type)
  - *Second* is the container (for now, just don't think about it)
  - *Third* is the class for comparator (the class that we use to compare)
    - This one is default to less<T>

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
    less<int> x;    // comparator that checks a < b
    greater<int> y; // checks a > b

    int a = 10;
    int b = 3;
    cout << x(a,b) << endl;    // 0
    cout << y(a,b) << endl;    // 1
}
```

### Using Comparator for set and map

- To use custom class with set and map, we need to do the same thing, let set and map know how to sort the data
  - Either make default ordering (overload <) in the custom class
  - Or use custom comparator when declare
- For set, the declaration is set<T, Compare = less<T>>
- For map, the declaration is map<Key, T, Compare = less<Key>>

**Assignment**

- Is any of `vector<int>`, `set<int>`, `map<int, string>`, `queue<bool>`, `stack<vector<int>>` comparable?
  - For any class that is “YES”, how it is ordered?
  - For example, if `vector<int>` is comparable, how `{1, 2, 3}` is compared to `{1, 2, 3, 4}` or `{2, 3, 4}`

**Answer**

- `vector<int>` compare each element from the left; `{2, 3, 4} > {1, 2, 3, 4} > {1, 2, 3}`
- `set<int>` compare the maximum element () of each set, if equal then look the next max
- `map<int, string>` compare by Key set, then compare max of value set
- `queue<bool>` is not comparable
- `stack<vector<int>>` is not comparable

**Summary****Data Structure Summary**

| Data Structure                       | Pro                                                                                                                                               | Cons                                                                                                                                                                                                                                         | Remark                                                                                                                                                                                                                      |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pair&lt;T1, T2&gt;</code>      | Nothings... It just a pair of two data type                                                                                                       |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                             |
| <code>vector&lt;T&gt;</code>         | <ul style="list-style-type: none"> <li>• Fast access [ ]</li> <li>• Fast append <math>\mathcal{O}(1)</math></li> </ul>                            | <ul style="list-style-type: none"> <li>• Slow find <math>\mathcal{O}(N)</math></li> <li>• Slow insert, Slow Erase <math>\mathcal{O}(N)</math></li> </ul>                                                                                     |                                                                                                                                                                                                                             |
| <code>set&lt;T&gt;</code>            | <ul style="list-style-type: none"> <li>• Fast find <math>\mathcal{O}(\log(N))</math></li> <li>• Item is sorted</li> </ul>                         | <ul style="list-style-type: none"> <li>• Slower to just append data than vector, stack, queue <math>\mathcal{O}(\log(N))</math></li> <li>• Iterate is also slow</li> <li>• Takes lots of memory <math>\mathcal{O}(\log(N))</math></li> </ul> | Require comparator                                                                                                                                                                                                          |
| <code>map&lt;Key, T&gt;</code>       |                                                                                                                                                   |                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Associative data type</li> <li>• Also require comparator of Key_Type</li> </ul>                                                                                                    |
| <code>stack&lt;T&gt;</code>          | <ul style="list-style-type: none"> <li>• Very fast push, pop <math>\mathcal{O}(1)</math></li> </ul>                                               | <ul style="list-style-type: none"> <li>• Very limited functionality but has special uses</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>• No iterator</li> <li>• Order of data coming out depends on something (stack, queue depends on WHEN it is pushed, pq depends on value)</li> <li>• pq requires comparator</li> </ul> |
| <code>queue&lt;T&gt;</code>          |                                                                                                                                                   |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                             |
| <code>priority_queue&lt;T&gt;</code> | <ul style="list-style-type: none"> <li>• Fast get max</li> <li>• Fast delete max</li> <li>• Data is sorted</li> <li>• Memory efficient</li> </ul> | <ul style="list-style-type: none"> <li>• Slower to just append data than vector, stack, queue</li> <li>• Very limited functionality</li> </ul>                                                                                               |                                                                                                                                                                                                                             |

**More Data Structure**

- C++ has more data structure not really covered right now
  - `list` is a vector with faster input / erase but does not have fast access
  - `unordered_set`, `unordered_map` are set and map that the data is not sorted but is much faster
  - `deque` is a queue that can push, pop at both ends
  - `multiset`, `multimap` are set and map that allows duplicated entries

**Lecture 8****Create Your Own CP::pair****Intro**

- We start writing our own data structure
  - In our own *namespace* (CP)
- We start with *pair*
  - Good introduction on writing a class

- Some C++ feature we need to use
  - const
  - pass-by-value, pass-by-ref
  - Header file

### What should we write in a class

- Class name and namespace
- Member *variables* which define what data we want to store in the class
- Member *functions* which define behavior of the class and how member variable is manipulated
  - Include lots of special function such as *constructor*, *destructor*, *overloading*

### CP:pair version 0.1 (minimal version)

- Templating
  - Parameter of a code
- No member functions
  - When we declare a pair, T1 and T2 must be declared as a type

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int, string> p1; // In p1, T1 is int, T2 is string
    CP::pair<bool, int> p2;  // In p2, T1 is bool, T2 is int
}
```

### Capabilities

- Can hold 2 pieces of data
- Can use template
- Works with operator =
- Has copy constructor
- Cannot check equal
- Cannot check less than

```
#include <iostream>
#include <string>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int, string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int, string> a(p1); // copy ctor
    p2 = p1;
}
```

```
    cout << p2.first << "," << p2.second << endl;
}
```

```
/*
if (p1 == p2) { // won't compile
    cout << "yes" << endl;
}
```

```
if (p1 < a) { // won't compile
    cout << "yes" << endl;
}
*/
```

### CP::pair version 0.2 (comparator overload)

```
#include <iostream>
#include <string>
#include <set>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // operator
        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        bool operator<(const pair<T1,T2>& other) const { // operator< must be const
            return ((first < other.first) ||
                    (first == other.first && second < other.second));
        }
    };
}
```

```
int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
    cout << p2.first << "," << p2.second << endl; // 20, somchai

    if (p1 == p2) {cout << "yes" << endl;} // yes
    if (p1 < a) {cout << "yes" << endl;}
    set<CP::pair<int,int>> s;
    s.insert({1,2});
    // Now we can compare and use less than (and also set, map, priority queue)
    cout << s.begin()->second << endl; // 2
}
```

## Const Member

### Const Parameter in Function

```
void test(int &x, const int& y) {
    x++;           // Okay; doable
    cout << y << endl; // Okay, we only read y
    for (int i=0; i<y; i++) { // Okay, too
        cout << i << endl;
    }
    y--; // ERROR: we promised NOT to change y
}
```

- Declared by putting a keyword const as a prefix
- Const parameter must not be modified inside the function
- Why? So that we know that the function does not modify the data
  - Especially when we pass-by-reference

### Difference of Pass-by-reference && Pass-by-value

#### Pass-by-value

- The *arguments* can be either constants or variables
- Modifying *parameters* (variable inside the function) does not change the *argument's* value
- The *argument's* value is copied to the parameter
  - **SLOWER!!!** Because we have to copy

#### Pass-by-reference

- The *arguments* must be variables
- The *parameters* are the argument's variables
  - Modify the *parameter* also modify the *argument's* variable
- **Faster**

### Const Member Function

```
class ccc {
public:
    int a,b;

    void inspect() const { // This function promises NOT to change anything
        if (a < b) cout << "yes" << endl; // Okay

        // b += 20; // <-- NOT OKAY
    }

    void mutate() { // This function might change something
        if (a < b) a += 10; // Okay
    }
};

void test2(ccc& changeable, const ccc& unchangeable) {
    changeable.inspect();
    changeable.mutate();
    unchangeable.inspect();
    unchangeable.mutate(); // ERROR: attempt to change unchangeable object
}
```

- Declared by putting a keyword const after the function declaration
- Const member function cannot modify any member data
  - Also cannot call any other function that is not const

- Why? So that we know that the function does not modify its data

## Constructor & List Initialization

### Custom Constructor

- In STL spec of `std::pair`, it has custom constructor called *initialization constructor*

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // Custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) {
            first = a;
            second = b;
        }
        // When we have a constructor, a default constructor is not auto-generated

        // Operator
        bool operator==(const pair<T1,T2> &other) {...}
        bool operator<(const pair<T1,T2>& other) {...}
    };
}

int main() {
    CP::pair<int, bool> p(10,false);
    CP::pair<string, int> q("abc",42), r("",0);

    cout << (q < r) << endl; // 0
    priority_queue<CP::pair<string,int>> pq;
    pq.push(r);
    pq.push(q);
    cout << pq.top().first << endl; // abc

    CP::pair<string, int> x(q);
    CP::pair<string, int> y = x;

    // All below cannot be compiled
    // CP::pair<string, int> w;
    // vector<CP::pair<int, int>> v(10);
}
```

### Initialization List

- Instead of writing a code to assign a value to each member, we can use *initialization list*
  - A little bit shorter code
  - Also little bit faster
  - Only way to init *const* member

```
namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;
```



```

        // custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

### Default Constructor

- A constructor used when we simply declare an object
- Auto-generated by init all members with its default constructor
- Won't be auto-gen if we have any other constructor

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
        T1 first;
        T2 second;

        // Constructor
        pair() : first(), second() { }
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

## Include & Header file

### Include File

- Usually, our data structure (which is written as a class) will be used by several programs in several files.
  - It is better NOT TO copy our data structure code into each file
- Rather, put our code in a file and include it where it is needed
  - Better if we want to change it
  - Better compilation
- Introducing ".h" files

### C++ header file (.h) and #include

- To put a content of one file into another file, we use `#include "filename"` keyword
- C++ will simply put the content of *filename* into where we `#include` it
- `#include` has more benefit
  - Separation of *declaration* (what it is) and *definition* (how it works)
  - Not really explored in this class
- We usually use .h for a file that we will include but that is not a rule, we can use other extension

### Problem with include

#### a.h

```

class a {
    int m1, m2;
};

```

#### b.h

```

#include "a.h"

```

```

class b {
    a x;
};

```

#### c.h

#### main.cpp (expanded)

```

class a {
    int m1, m2;
};

```

```

class b {
    a x;
};

```

```

// There are multiple copies of class a
class a {
    int m1, m2;
};

```

```
#include "a.h"
```

```
class c {
    a y;
};
```

#### main.cpp

```
#include "b.h"
#include "c.h"
```

```
int main() {
    b b1;
    c c1;
}
```

#### Fix problem

##### a.h

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif
```

##### b.h

```
#include "a.h"
```

```
class b {
    a x;
};
```

##### c.h

```
#include "a.h"
```

```
class c {
    a y;
};
```

```
class c {
    a y;
};
```

```
int main() {
    b b1;
    c c1;
}
```

#### main.cpp (expanded)

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif

class b {
    a x;
};

// class a here is taken out of
// compilation because A_H is defined
#ifndef A_H
// #define A_H
// class a {
//     int m1, m2;
// };
#endif

class c {
    a y;
};

int main() {
    b b1;
    c c1;
}
```

#### Summary

##### Major

- **Templating**: allow user to write a code that works with different data type
- **Constructor**: a function called when we create a variable
  - Default constructor
- **Operator overloading**: for less-than and equality
- pass-by-ref vs pass-by-value

##### Minor

- Header file
- const
- List initialization

### Exercise (no grader)

1. Modify operator< so that it compare *second* before *first*
2. Modify operator< so that when we call sort(v.begin(), v.end()) where w is a vector of our pair, it is sorted from *Max* to *Min*
3. Write operator!= and operator>=

### Final version CP.h

```
#ifndef _CP_PAIR_INCLUDED_
#define _CP_PAIR_INCLUDED_

namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;

        pair() : first(), second() {}

        pair(const T1 &a, const T2 &b) : first(a), second(b) { }

        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        // 1.
        bool operator<(const pair<T1,T2>& other) const {
            return ((second < other.second) ||
                    (second == other.second && first < other.first));
        }

        // 2.
        bool operator<(const pair<T1,T2>& other) const {
            return ((first > other.first) ||
                    (first == other.first && second > other.second));
        }

        // 3.
        bool operator!=(const pair<T1,T2>& other) const {
            return !(*this == other);
        }

        bool operator>=(const pair<T1,T2>& other) const {
            return !(*this < other);
        }

    };
}
#endif
```

## Lecture 9

### CP::vector

Our first “real” data structure

#### Intro

- Now we will create more complex data structure `CP::vector`
- It can store variable length array
  - Implemented as a dynamic array
- Can be accessed by operator `[]`
  - Additional operator to be overloaded
- We also have to create our own iterator
  - Implemented as a pointer

#### Key Idea

- Vector stored 3 things (3 member data)
  - `mData`: A dynamic array, large enough space to store current data and might have reserved space
  - `mSize`: Number of data stored
  - `mCap`: Size of the dynamic array (maybe large than `mSize`)
- If the dynamic array is full and more data is being added, we create a new dynamic array and relocate data to the new array
  - This is called **expand**
  - Each **expansion** takes very long time
- Dilemma
  - Large reserve = less often relocation but use more memory
  - Small reserve = more frequent relocation but less memory

#### Example

```
mSize  mCap      mData
  4      5    [10, 3, 1, 5, ]
```

```
push_back(9)
```

```
  5      5    [10, 3, 1, 5, 9]
```

```
push_back(4)
```

```
  6     10    [10, 3, 1, 5, 9, 4, , , , ]
```

#### How much reserve should we have?

- Whenever we need to relocate, we **double** the capacity that we currently have
- If we start with a vector with zero size and continuously add data one by one, for example by `push_back`
  - Each expansion will take time **equal** to **the number of data** when we relocate (this is very slow)
- By doubling the size every time we expand, we can show that on average, **each addition of a data** (such as `push_back`, `insert`) **takes constant time!!!**

### Pointer

How to implement a dynamic array (and also iterator)

#### Pointer & Memory

- Each variable is some block in computer memory
  - Programming language just map our **variable name** to that **block of memory**
  - Programming language works with the address of that block
- Pointer variable is a variable that stor **address of memory**

- ▶ Pointer needs type i.e., address of int is not the same as address of bool
- ▶ We can use operator & to ask for the **address** of a **variable**
- ▶ We can use operator \* to ask for the **data** of an **address**

### Example

```
int x, y;    // this is normal variable x and y
int *a;     // this is int pointer variable a
int *b;     // another int pointer

x = 10;     // say x is at address 3267
a = &x;     // a points to x
b = a;      // b also points to x
*b = 30;    // change the address value to 30
y = *b;     // set value of y to 30 (hard copy)

cout << &x << endl;    // 0x7ffee22885ac
cout << &y << endl;    // 0x7ffee22885a8 +4
cout << &a << endl;    // 0x7ffee22885a0 +8
cout << &b << endl;    // 0x7ffee2288598 +8
cout << sizeof(int) << endl;    // 4 bytes
cout << sizeof(*int) << endl;    // 8 bytes
```

### Pointer Arithmetic

- Pointer can be added, subtracted by integer
  - ▶ It **moves the address by the size of the type** of the pointer
  - ▶ For example, when X is an int (which is 4 bytes) X + 10 result in an address 40 bytes away from X
- Two pointers of the same type can be subtracted
  - ▶ The result is the address **difference divided by size of the type** of the pointer

```
int main() {
    bool x, y, z;
    x = 1; y = 2; z = 3;
    bool *a, *b;

    a = &x;
    b = a+2;

    cout << "&x = " << &x << endl;    // &x = 0x6afed4
    cout << "&y = " << &y << endl;    // &y = 0x6afed8
    cout << "&z = " << &z << endl;    // &z = 0x6afedc
    cout << " a = " << a << endl;    // a = 0x6afed4
    cout << " b = " << b << endl;    // b = 0x6afedc
    cout << b-a << endl;            // 2
}
```

### Dynamic Array

- **Dynamic Array** variable of type T is a **pointer** to the **starting address** of consecutive block of type T
- Let A be a dynamic array of int
  - ▶ A[x] refer to the  $x^{\text{th}}$  block starting from A
- Static array works in the same way, that's why accessing A[x] is very fast for an array
  - ▶ It just refers to the address A[x] is  $*(A + x * \text{sizeof}(T))$

### new and delete operator

- For a datatype T, new T *allocates* a block with a size of T and then call a *constructor* of T, return the address of that memory

- For a datatype T, `new T[n]` allocates n blocks of T and then call a *constructor* of T in each block, return the address of the first block
  - For example, we use `new int [10]` to create a dynamic int array of 10 elements
- For a pointer X, `delete X` calls the *destructor* and then *de-allocates* memory pointed by X
- For a dynamic array X, `delete []` calls the *destructor* of all blocks in the dynamic array and *de-allocates* the memory allocated by X

### Example

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    test *a, *b;    // pointers of test

    a = new test;    // created new test pointed by a
    a->data = 10;    // (*a).data = 10
    cout << a->data << endl;    // 10
    delete a;        // destroyed

    b = new test[4]; // created 4x
    b[0].data = 10;
    b[1].data = 20;
    b[2].data = 30;
    b[3].data = 40;
    delete [] b;     // deleted 4x
}
```

### Memory Leak

- For everything, this is created by new, we must call delete on it
- If you do not, that memory is not deleted until all memory is used up

```
#include <iostream>
using namespace std;

void leaked() {
    int *a;
    a = new int[2000];
}

int main() {
    for (int i=0; i<1000000; i++) {
        int *a;
        cout << i << endl;
        leaked();
    }
}

// terminate called after throwing an instance of 'std::bad_alloc'
```

### Smart Pointer (NOT A SUBJECT OF THIS COURSE)

- A better way is to use C++ smart pointer `shared_ptr(T)`
- Smart pointer is a pointer that can delete itself when it go out of scope

- Similar concept to [Java Garbage Collection](#)
- Still possible to have memory leak

## vector.h

### Version 0.1

- Start with vector that can do push\_back, pop\_back, and []
- Also with custom constructor

```
namespace CP {
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {...}
    void expand(size_t capacity) {...}
    void ensureCapacity(size_t capacity) {...}
public:
    vector() {...}
    vector(size_t capacity) {...}
    ~vector() {...}
    // access
    T& at(int index) {...}
    T& operator[](int index) {...}
    // modifier
    void push_back(const T& element) {...}
    void pop_back() {...}
};
}
```

### Basic Constructor

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

public:
    vector() {
        int cap = 1;
        mData = new T[cap]();
        mCap = cap;
        mSize = 0;
    }

    vector(size_t cap) {
        mData = new T[cap]();
        mCap = cap;
        mSize = cap;
    }
}
```

```
vector<int> v;
```

```
vector<int> w(5);
```

|       |      |       |  |       |      |             |
|-------|------|-------|--|-------|------|-------------|
| mSize | mCap | mData |  | mSize | mCap | mData       |
| 0     | 1    | [ ]   |  | 5     | 5    | [0,0,0,0,0] |

### Destructor

- Since we new mData, we have to delete it
  - Or face a memory leak problem

```
~vector() {
    delete [] mData;
}
};
```

### Object Life Cycle

- Normal object
  - Object is created (constructor called) when declared
  - Object is destroyed (destructor called) when go out of scope
- Object created by new (both new T or new T[]) Object is created when new Object is destroyed when delete

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    cout << "-- Life cycle --" << endl;
    cout << "- normal cycle -" << endl;
    test u;
    u.data = 99;
    for (int i=0; i<5; i++) {
        test t;
        t.data = i*10;
    }
}
```

### Output

```
-- Life cycle --
- normal cycle -
created      <-- u
created
0 destroyed
created
10 destroyed
created
20 destroyed
created
30 destroyed
created
40 destroyed
99 destroyed <-- u
```



### Accessing Data

- The return type is T& which is a reference
- Same deal as pass-by-reference, this is called **return-by-reference**
  - So we can do `v[i] = 30` or `v[i]++`
- What is returned is actually that variable
- Also notice the difference between `at()` and `operator[]`

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {
        if (n < 0 || (size_t)n >= mSize) {
            throw std::out_of_range("index out of range");
        }
    }

public:
    T& at(int index) {
        rangeCheck(index);
        return mData[index];
    }

    T& operator[](int index) {
        return mData[index]; // return the variable
    }
};
```

### Add, remove data

- `push_back` first check if we have reserved space
  - If not, we expand
- Then, the data is put to `mData[mSize]`
- Removing Data is done by just reduce the size

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    void expand(size_t capacity) {
        T *arr = new T[capacity](); // create a new dynamic array
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[i]; // move all data
        }
        delete [] mData;
        mData = arr; // delete old data and point to new one
        mCap = capacity;
    }

    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap; // double the size
            expand(s);
        }
    }
};
```

```

    }
}
public:
    void push_back(const T& element) {
        ensureCapacity(mSize+1);
        mData[mSize++] = element; // add size and put element
    }
    void pop_back() {
        mSize--;
    }
};

```

## Vector Constructors

### Problem of vo.1

- Copy constructor and assignment operator is incorrect
  - It is auto generate to copy all variables (but not the data it points to)
- Rule of three in c++
  - Consider destructor, copy constructor, assignment operator
  - If any of them is written in the code, we mostly need all of them
- Since c++11, it's rule of four an a half

```

int main() {
    CP::vector<int> w(5);

    for (int i=0; i<5; i++) w[i] = i*10;
    CP::vector<int> x(w); // this creates shallow copy of w
    CP::vector<int> y = w;
    x[3] = -1; // changes both w and y
    cout << y[3] << endl; // -1
    cout << w[3] << endl; // -1
}

```

### vo.2, add small access functions

- empty and size also exists in other data structure
- size\_t is non-negative integer type

```

bool empty() const {
    return mSize == 0;
}

size_t size() const {
    return mSize;
}

size_t capacity() const {
    return mCap;
}

```

### vo.2 adding copy constructor & assignment operator

```

// copy constructor
vector(const vector<T>& a) {
    mData = new T[a.capacity()]();
    mCap = a.capacity(); // copies only value
    mSize = a.size();
    for (size_t i=0; i<a.size(); i++) {
        mData[i] = a[i];
    }
}

```

```

    }
}

// copy assignment operator
vector<T>& operator=(vector<T> &other) {
    // protect against self-destruct
    if (mData != other.mData) {
        // delete current data
        delete [] mData;
        // copy the new data
        mData = new T[other.capacity()]();
        mCap = other.capacity();
        mSize = other.size();
        for (size_t i=0; i<a.size(); i++) {
            mData[i] = a[i];
        }
    }
}

```

### Copy-and-swap

- Utilize written copy-constructor and destructor
- Shorter code

```

// copy assignment operator using copy-and-swap idiom
vector<T>& operator=(vector<T> other) { // notice the pass-by-value!!!
    // other is copy-constructed which will be destruct at the end of this scope
    // we swap the content of this class to the other class and let it be destructed
    using std::swap;
    swap(this->mSize, other.mSize);
    swap(this->mCap, other.mCap);
    swap(this->mData, other.mData);
    return *this;
}

```

### Iterator, Insert, Erase, & Analysis

#### vo.3 Iterator and typedef keyword

```

template <typename T>
class vector {

protected:
    T *mData;
    size_t mCap;
    size_t mSize;
protected:
    typedef T* iterator; // let iterator be type name of T pointer

    // iterator
    iterator begin() {
        return &mData[0];
    }

    iterator end() {
        return begin()+mSize;
    }
}

```

- See that pointer works just like how `std::vector::iterator` works
- In fact, iterator is actually a pointer
- `typedef` keyword allows us to map a type name
  - `CP::vector<int>::iterator` is `int*`
  - `CP::vector<bool>::iterator` is `bool*`

### insert

- `push_back` actually call `insert(end(), element)`
- Question: why we need `pos`?

```
iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize+1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1];
    }
    mData[pos] = element;
    mSize++;
    return begin()+pos;
}
```

```
void push_back(const T& element) {
    insert(end(), element);
}
```

### erase

- See that both `insert` and `erase` also change `mSize`

```
void erase(iterator it) {
    while((it+1)!=end()) {
        *it = *(it+1);
        it++;
    }
    mSize--;
}
```

### Exercise

- Read the following function and see how it works in `vector.h`
  - `resize`, `clear`
  - non-stl function
    - `insert_by_pos`
    - `erase_by_pos`
    - `erase_by_value`
    - `contains`
    - `index_of`
- Read in [here](#)

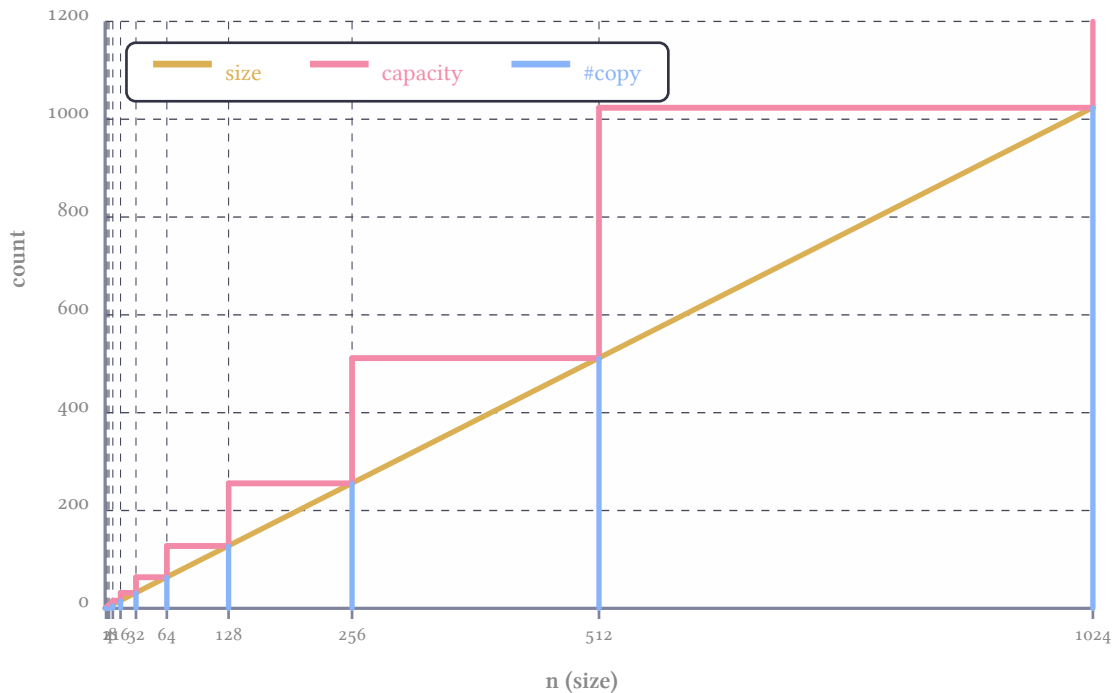
Will do this part later

### Analysis of how many data is copied by `push_back`

- When full, `push_back` have to move all data to a new dynamic array
- `ensureCapacity` double the size

### Size and Capa & Copy Count

- How much copy we need?



## Lecture 10

### Stack

#### Intro

- Now we will create less complex data structure `CP::stack`
- Just like a vector without iterator, insert, erase, resize, at end operator[]
  - Add `top()` which is just a shorthand of looking at the last element
- That's it, really

#### Key Idea

- The data is stored in the same way as a vector
  - The first element of `mData` is the bottom of stack while the last element is the top of the stack
- We just take `vector.h` and remove unnecessary function

```

| | stack
| |
| A | mSize mCap mData
| X | 4 5 [A, B, X, A]
| B |
| A |
|__|

```

#### stack.h

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        void expand(size_t capacity) {...}
    };
}

```

```

    void ensureCapacity(size_t capacity) {...}
public:
    // constructor
    stack(const stack<T>& a) {...}
    stack() {...}
    stack<T>& operator= {...}
    ~stack() {...}
    // capacity operation
    bool empty() const {...}
    size_t size() const {...}
    // access
    const T& top() const {
        return mData[mSize-1];
    }
    // modifier
    void push(const T& operator) {...} // This is push_back
    void pop() {...} // This is pop_back
};
}

```

### Speed of each operation

- All read operation always take constant time
  - `size()`, `top()` simply return something that is directly accessible
- All modify operation also take constant time
  - `push()` is constant on average (same as `push_back` of vector)
  - `pop()` is always constant

### Stack by Vector

- Instead of writing our own function, there is another way to write a stack
- We simply use `vector` as our sole data member
- Benefit: `code reuse`
- Drawback: `almost none` except that we need one more layer of function call

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        vector<T> v;
    public:
        // default constructor
        stack() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& top() const { return v.back(); }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.pop_back(); }
    };
}

```

## Lecture 11

### CP::queue

Will the circle be unbroken?

#### Intro

- Queue, unlike stack, require more sophisticated technique to achieve fast performance
- We start by writing a simple class that just work (slowly)
- Then we try to improve it

#### Key Idea

- Just like stack, we will use the same format as vector, using dynamic array to store data
- However we have to somehow manage how we works with `front()` and `back()` of the queue

#### vo.1 simple implementation of the queue

- To illustrate this idea, we will use a **vector** as our data member
- `push(e)` is simply `v.push_back(e)`, this is fast
- `front()` is `v[0]`, `back()` is `v[v.size()-1]`, this is also fast
- `pop()` is `v.erase(v.begin())`, this is slow (always proportional to `v.size()`)
  - Unlike `std::queue` which has very fast `pop()`

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
    public:
        // default constructor
        queue() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& front() const { return v[0]; }
        const T& back() const { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.erase(v.begin()); }
    };
}
```

#### vo.1 example

```
CP::queue<int> q;      q  v
                        mSize mCap mData
                        0    1    ----> [ ]

q.push(10);            q  v
                        mSize mCap mData
                        1    1    ----> [10]

q.push(20);            q  v
                        mSize mCap mData
                        2    2    ----> [10, 20]

q.push(30);            q  v
                        mSize mCap mData
                        3    4    ----> [10, 20, 30, ]

q.pop();               q  v
                        /    /
```

|       |      |       |    |              |
|-------|------|-------|----|--------------|
| mSize | mCap | mData | /  | /            |
| 2     | 4    | ----  | -> | [20, 30, , ] |

**vo.2 faster queue**

- Add more data member mFront, initialized as 0
- push(e) is simply v.push\_back(e), this is fast
- front() is v[mFront], back() is v[v.size()-1], this is also fast
- pop() is mFront++, this is fast
  - However, we don't really remove anything when pop

```
#include <vector>
```

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
        int mFront;
    public:
        // constructor
        queue() : v(), mFront() {}
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size() - mFront; }
        // access
        const T& front() const { return v[mFront]; }
        const T& back() { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { mFront++; }
    };
}
```

**vo.2 example**

|                   |        |       |                        |
|-------------------|--------|-------|------------------------|
| CP::queue<int> q; | q      | v     |                        |
|                   | mFront | mSize | mCap mData             |
|                   | 0      | 0     | 1 ----> [ ]            |
| q.push(10);       | q      | v     |                        |
|                   | mFront | mSize | mCap mData             |
|                   | 0      | 1     | 1 ----> [10]           |
| q.push(20);       | q      | v     |                        |
|                   | mFront | mSize | mCap mData             |
|                   | 0      | 2     | 2 ----> [10, 20]       |
| q.push(30);       | q      | v     |                        |
|                   | mFront | mSize | mCap mData             |
|                   | 0      | 3     | 4 ----> [10, 20, 30, ] |
| q.pop();          | q      | v     |                        |
|                   | mFront | mSize | mCap mData             |
|                   | 1      | 3     | 4 ----> [10, 20, 30, ] |

**Problem with vo.2**

- Fast but **use too many space**
- Queue grows according to **how many time push is called**
  - regardless of how many pop is called
- The data stored in the vector can be **much larger** than the actual data in the queue
- Does not really work in real world



```
for (int i=0; i<1000000; i++) {
    q.push(i);
    q.pop();
}
std::cout << q.size() << std::endl;
```

## Circular Queue

### Final Idea

| mFront      | mSize | mCap | mData                           | *       |
|-------------|-------|------|---------------------------------|---------|
| 2           | 2     | 4    | ----> [ , , 30, 40]             |         |
| q.push(50); |       |      |                                 |         |
| mFront      | mSize | mCap | mData                           | *       |
| 2           | 3     | 4    | ----> [50, , 30, 40]            |         |
| q.push(60); |       |      |                                 |         |
| mFront      | mSize | mCap | mData                           | *       |
| 2           | 4     | 4    | ----> [50, 60, 30, 40]          |         |
| q.pop();    |       |      |                                 |         |
| mFront      | mSize | mCap | mData                           | *       |
| 3           | 3     | 4    | ----> [50, 60, , 40]            |         |
| q.push(99); |       |      |                                 |         |
| mFront      | mSize | mCap | mData                           | *       |
| 3           | 4     | 4    | ----> [50, 60, 99, 40]          |         |
| q.push(1);  |       |      |                                 |         |
| mFront      | mSize | mCap | mData                           | * \ \ \ |
| 0           | 5     | 4    | ----> [40, 50, 60, 99, 1, , , ] |         |

- We take vo.2 and **reuse** the area at the beginning of mData
  - Expand when necessary
  - Re-arrange when expand

### Circular Queue

- We can think of mData to be circular
  - End of the last element of the mData is connected to the first element
- Consider  $i^{\text{th}}$  element
  - the next element is  $(i+1) \% \text{mCap}$
  - The previous element is  $(i-1+\text{mCap}) \% \text{mCap}$
  - Next k element is  $(i+k) \% \text{mCap}$

## Circular Queue Implementation

### queue.h

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        size_t mFront;
        void expand(size_t capacity) {...}
        void ensureCapacity(size_t capacity) {...}
    public:
        // constructor - almost the same but have to take care of mFront
        queue(const queue<T>& a) {...}
        queue() {...}
    };
}
```

```

    queue<T>& operator=(queue<T> other) {...}
    ~queue() {...}
    // capacity function - same as vector
    bool empty() const {...}
    size_t size() const {...}
    // access - circular queue implementation
    const T& front() const {...}
    const T& back() const {...}
    // modifier
    void push(const T& element) {...}
    void pop() {...}
};
}

```

### Ctor, Dtor, copy

```

template <typename T>
class queue {
protected:
    T *mData;   size_t mCap;   size_t mSize;   size_t mFront;
public:
    // default constructor
    queue() : mData(new T[1]()), mCap(1),
             mSize(0), mFront(0) { }
    // copy constructor
    queue(const queue<T>& a) : mData(new T[a.mCap]()), mCap(a.mCap),
                             mSize(a.mSize), mFront(a.mFront) {
        for (size_t i = 0; i < a.mCap; i++) {
            mData[i] = a.mData[i];
        }
    }
    // copy assignment operator
    queue<T>& operator=(queue<T> other) {
        using std::swap;
        swap(mSize, other.mSize);
        swap(mCap, other.mCap);
        swap(mData, other.mData);
        swap(mFront, other.mFront);
        return *this;
    }
    ~queue() {
        delete [] mData;
    }
};

```

### front(), back(), pop()

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
public:
    // access
    const T& front() const {
        return mData[mFront];
    }
};

```

```

    }
    const T& back() const {
        return mData[(mFront + mSize - 1) % mCap];
    }
    // modify
    void pop() {
        mFront = (mFront + 1) % mCap;
        mSize--;
    }
};

```

- back = mFront + mSize - 1
  - Also circular (by % mCap)
- pop = move mFront by 1
  - Also circular
  - Also change size

### push, expand

- push add data to (mFront + mSize) % mCap
  - The space just after back()
- Expand re-pack the mData so that mFront is 0
- ensureCapacity is the same

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
    void expand(size_t capacity) {
        T *arr = new T[capacity]();
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[(mFront + i) % mCap];
        }
        delete [] mData;
        mData = arr;
        mCap = capacity;
        mFront = 0;
    }
    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
            expand(s);
        }
    }
public:
    void push(const T& element) {
        ensureCapacity(mSize+1);
        mData[(mFront + mSize) % mCap] = element;
        mSize++;
    }
};

```

### Analysis

- All access, modification is fast (constant time)

- Space is re-used
  - It is not shrunk when mSize reduce
  - Space is not more than double of maximum mSize during its lifetime

### Queue Exercise

- We implement circular queue by maintain mFront and use circular logic ( $\% \text{ mCap}$ ) to calculate the position of back of the queue
  - Can we maintain mBack instead?
  - Can we maintain both mFront and mBack but not mSize?
- How about mCap, if we know mFront, mSize, mBack, can we calculate mCap?

| mFront | mSize | mBack | front()            | back()                                              | size()           |
|--------|-------|-------|--------------------|-----------------------------------------------------|------------------|
| YES    | YES   | No    | $v[\text{mFront}]$ | $v[(\text{mFront} + \text{mSize}) \% \text{ mCap}]$ | $\text{mSize}()$ |
| No     | YES   | YES   | ????               | $v[\text{mBack}]$                                   | mSize            |
| YES    | No    | YES   | $v[\text{mFront}]$ | $v[\text{mBack}]$                                   | ????             |

### Answer

- Use  $\text{front}() = v[(\text{mCap} + \text{mBack} - \text{mSize}) \% \text{ mCap}]$
- Use  $\text{size}() = (\text{mCap} + \text{mBack} - \text{mFront}) \% \text{ mCap}$ , but this will break when  $\text{mBack} == \text{mFront}$  because it can't determine whether the queue is empty or full.
- Impossible, mCap can be scaled to any size larger than the current elements.

### Now, meet deque

- Can you modify queue to include
  - $\text{push\_front}()$ , add to the front of the queue
  - $\text{pop\_back}()$ , remove from back of the queue
- All operation should still be constant time

### Answer

```
void push_front(const T& element) {
    ensureCapacity(mSize+1);
    mFront = (mCap + mFront - 1) % mCap;
    mData[mFront] = element;
    mSize++;
}

void pop_back() {
    mSize--;
}
```

## Lecture 12

### Complexity Analysis

How fast is our code?

#### Overview

- Introducing a **measurement** of efficiency of a program
- Why we need measurement
- How we measure
  - How to analyze our code to get a measurement
- What is measurement
  - **Asymptotic Notation**

### Key Idea

- We need a useful way to describe efficiency of our code
  - Useful = able to be **easily use to predict** how much resource (time / memory) that our program will need
  - Useful = **not overly complex** in analysis
  - Need to balance between usefulness and complexity
- Ultimately, we introduce a **class of efficiency** that says how our code use resource with respect to size of data
  - Focus on **on growth of resource usage**

### Preview

```
int find_max(vector<int> v) {
    int m = v[0];
    for (size_t i = 0; i < v.size(); i++)
        if (v[i] > m)
            m = v[i];
    return m;
}
```

- This code takes time **directly proportional** to **the size of the data**
  - Size  $N$  takes time  $T$
  - Size  $5N$  should take time  $5T$

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

- This code takes time **directly proportional** to **the square of the size of the data**
  - Size  $N$  takes time  $T$
  - Size  $5N$  should take time  $25T$

### Why don't we use real world clock?

- Ultimately, we want to know **how long our program takes** to do each operation
  - Use in design, how much resource we need
  - Help use **choose appropriate data structure**
- **Real world clock measurement is possible** but has **many drawback**
  - System dependency
  - Too complex (we have to build our system)
  - Too specific

### Dependency of System

- Same program in different system = different time
  - CPU
  - RAM
  - Compiler
  - Operating parameter
    - Heat? Other running program?

### How to measure

- Counting instruction
  - Depend on code only
- Dependency on size of data
  - Focus on large data

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

// 1  
 //  $2 + n + n$  ( $n = v.size()$ )  
 //  $(2 + n + n) * n$   
 //  $2 * n * n$   
 //  $1 * ????$  ( $\leq (d)$ )  
 // 1

```
}
```

```
// Total = 1 + 2+n+n + (2+n+n)n + 2n^2 + n^2 + 1
//          = 4 + 4n + 5n^2
```

### Time per instruction

- In reality, different CPU instructions use different time
- Same instruction but different CPU also use different number of cycle
- **However, we just ignore it**
  - For now

### Focusing on large data

- In many cases, the size of the data we are working with will affect the time our code use
- Large data usually mean longer time
- What matter is when the data is large

### Growth rate simplifies analysis

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = i+1; j < v.size(); j++)
            if (v[i] + v[j] == k)
                count++;
    return count;
}
```

// i=0: 1 + n-0 + n-1  
 // i=1: 1 + n-1 + n-2  
 // <-- i=2: 1 + n-2 + n-3  
 // . . .  
 // i=n-1: 1 + 1 + 0  
 // sum =  $\Sigma(1+2n-2i-1) = n^2 + n$

```
// Total = 1 + 2+n+n + n^2+n + n^2 + n^2 + 1
//          = 4 + 3n + 3n^2
```

### Small Detail

| n      | $5n^2 + 4n + 4$    |          | $3n^2 + 3n + 4$    |          | $n^2$              |        |
|--------|--------------------|----------|--------------------|----------|--------------------|--------|
|        | raw                | growth   | raw                | growth   | raw                | growth |
| 10     | 544                |          | 334                |          | 100                |        |
| 20     | 2084               | 3.830 88 | 1264               | 3.784 43 | 400                | 4      |
| 40     | 8164               | 3.917 47 | 4924               | 3.895 57 | 1600               | 4      |
| 80     | 32 324             | 3.959 33 | 19 444             | 3.948 82 | 6400               | 4      |
| 160    | 128 644            | 3.979 83 | 77 284             | 3.9747   | 25 600             | 4      |
| 320    | 513 284            | 3.989 96 | 308 164            | 3.987 42 | 102 400            | 4      |
| 640    | 2 050 564          | 3.994 99 | 1 230 724          | 3.993 73 | 409 600            | 4      |
| 1280   | 8 197 124          | 3.9975   | 4 919 044          | 3.996 87 | 1 638 400          | 4      |
| 2560   | 32 778 244         | 3.998 75 | 19 668 484         | 3.998 44 | 6 553 600          | 4      |
| 5120   | $1.31 \times 10^8$ | 3.999 37 | 78 658 564         | 3.999 22 | 26 214 400         | 4      |
| 10 240 | $5.24 \times 10^8$ | 3.999 69 | $3.15 \times 10^8$ | 3.999 61 | $1.05 \times 10^8$ | 4      |
| 20 480 | $2.1 \times 10^9$  | 3.999 84 | $1.26 \times 10^9$ | 3.9998   | $4.19 \times 10^8$ | 4      |

When counting instruction, it is usually OK to focus on **most executed line**

### Measurement by Growth Rate

What?

Why?

- Growth rate = how much resource usage growth with respect to change of input
  - Resource usage = number of instruction used
  - Input = size of data
- Emphasizes long term trend
- System independent
  - The result can be used to predict behavior on any system
- Focus on change of resource usage with respect to size of input
- Can regard small detail
  - Simple to calculate
  - Applicable in real world

## Asymtotic Notation

Classification of growth rate

### Overview

- Formally, it is a **set of function** having the **growth rate** related to something
- The definition focus **on growth of the function** while **disregard small detail**
- Also provide some workaround on dependency of value of input

### What is?

- A kind notation written as  $O(f(n))$ 
  - $O$  can be one of  $\mathcal{O}, \Theta, \Omega, \sigma, \omega$
  - $f(n)$  is some expression
- Example  $\mathcal{O}(n)$  or  $\Theta(n^2 + 3)$  or  $\omega(n^2 \log(n))$
- Usage
  - “This code is  $\mathcal{O}(n)$ ” (read as Big-Oh of n)
  - “This function takes time in  $\Theta(\log(n))$ ” (read as Big-theta of log n)
  - “Time complexity of this program is  $\mathcal{O}(n^2)$ ”

### Meaning

- “A is  $\Theta(f(x))$ ” means the **growth rate of A** is **equal** to the **growth rate of  $f(x)$**
- “B is  $\mathcal{O}(f(x))$ ” means the **growth rate of B** is **less than or equal** to the **growth rate of  $f(x)$**
- For  $\Omega, \sigma, \omega$ , (**Big-Omega**, **little-oh**, **little-omega**), we won’t use it for now but the meaning is similar (which are **more than or equal**, **less than**, **more than**, respectively)
- Conventionally, we usually use  **$N$  for the size of the data**

### Usage

- Let  **$f(n)$**  be the number of instruction needed by **code A** when the **size of data is  $n$**
- We will calculate asymptotic notation that  **$f(n)$**  is a member of
  - Find  $g(n)$  such that  $f(n)$  is  $\mathcal{O}(g(n))$  or  $\Theta(g(n))$  (or  $\Omega(g(n))$  or ...)
- Let’s say we have analyze that  **$f(n)$  is  $\mathcal{O}(g(n))$** 
  - We now understand that the growth rate of instruction required by  $n$  grows slower or the same as how  $g(n)$  grow

### Comparing growth rate of $f(n)$ and $g(n)$

- The relation of growth rate of  $f(n)$  and  $g(n)$  depends on the value of  $f(n)/g(n)$  when  $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \begin{cases} 0 & \text{if } f(n) \text{ grows slower than } g(n) \\ c & \text{if } f(n) \text{ grows similarly to } g(n) \\ \infty & \text{if } f(n) \text{ grows faster than } g(n) \end{cases}$$

### Example

- $f(n) = 4 + 3n + 4n^2$
- $g(n) = n^2$

$$\lim_{n \rightarrow \infty} \frac{4n^2 + 3n + 4}{n^2} = \lim_{n \rightarrow \infty} \left( 4 + \frac{3}{n} + \frac{4}{n^2} \right) = 4$$

- Hence  $f(n)$  grows similarly to  $g(n)$ 
  - Therefore,  $f(n) = \Theta(n^2)$

#### Another Example

- $f(n) = 0.00005n^2$
- $g(n) = 100000n$

$$\lim_{n \rightarrow \infty} \frac{0.00005n^2}{100000n} = \lim_{n \rightarrow \infty} 10^{-10}n = \infty$$

- Hence  $f(n)$  grows faster than  $g(n)$  (also means  $g(n)$  grows slower than  $f(n)$ )
  - Therefore  $g(n) = \mathcal{O}(n^2)$
  - and  $f(n) = \Omega(100000n)$

#### Dependency on the value of input

- Consider `vector::insert(iterator it, T value)`
- The time it takes depends on both the size of the vector and the value of it
  - Larger size  $\rightarrow$  more time
  - Closer to `end()`  $\rightarrow$  less time

#### Big-O describes upper bound

- `vector::insert` is  $\mathcal{O}(n)$ 
  - Its growth rate does not exceed  $n$
  - There are cases that it may grow less than  $n$  (insert at `end()`)
- This is very useful in real world
  - Knowing maximum load
  - Not overly complex in analysis

#### Big-O Example

- Find is  $\mathcal{O}(n)$ 
  - At worst, it can't find value and a & b points to `begin()` & `end()`
    - This case, find growth rate is  $n$
  - At best, it always find value at the first position (a)
    - This case, find growth as 1

```
bool find(iterator a, iterator b, T value) {
    while (a < b) {
        if (*a == value)
            return true;
        a++;
    }
    return false;
}
```

#### L'Hôpital's Rule

$$\lim_{n \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{n \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

If  $f(x), g(x)$  must be diffable,  $g(x)$  is non-zero,  $\lim_{n \rightarrow \infty} f'/g'$  exists

- Can help
- $F(n) = \log n$
- $g(n) = n^{0.5}$



$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{\log(n)}{\sqrt{n}} &= \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(10)\sqrt{n}} \\
&= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\ln(n)}{\sqrt{n}} \\
&= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} \\
&= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n}} \\
&= 0
\end{aligned}$$

**Exercise**

- $f(n) = \lg^c n$
- $g(n) = n^k$
- We know that  $c > 0, k > 0$
- Does  $f(n)$  grow slower than  $g(n)$ ?

Will do this part later

**Big Theta****Big-Theta is tight bound**

- `std::count` always go through entire array
- Regardless of the value in the array, it always perform `if (*first == value)`

```

size_t count(iterator first, iterator last, const T& value) {
    size_t ret = 0;           // 1
    for (; first != last; ++first) { // n + n
        if (*first == value)    // 1 * n
            ret++;              // 1 * ??? ( <= (d) )
    }
    return ret;               // 1
}

```

```

// Total = 1 + n+n + 1*n + 1*n + 1
//         = 2 + 4n

```

**More Example**

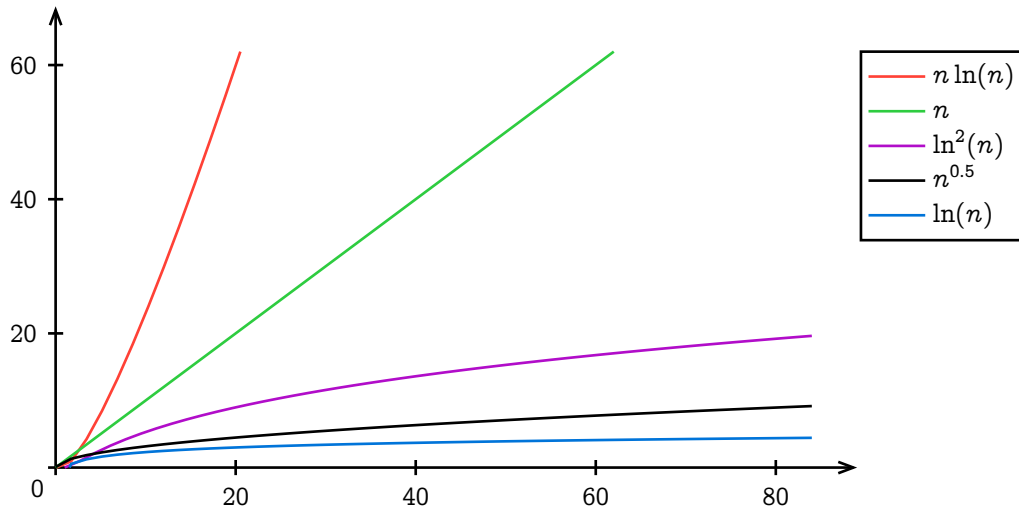
| $\Theta(1)$ | $\Theta(n)$                  | $\Theta(n^2)$                            |
|-------------|------------------------------|------------------------------------------|
| $f(n) = 5$  | $f(n) = n$                   | $f(n) = n^2$                             |
| $f(n) = 0$  | $f(n) = n + 3$               | $f(n) = \binom{n}{2} = \frac{n(n-1)}{2}$ |
| $f(n) = c$  | $f(n) = \frac{n}{1200} + 86$ | $f(n) = 400n^2 + an + b$                 |
|             | $f(n) = 40000000n$           |                                          |

Observation: multiplicative and addition constants in  $f(n)$  can usually be ignored, since it will be disregarded by  $\lim$ , Degree cannot.

**Well known growth rate class**

|                               |               |
|-------------------------------|---------------|
| $\Theta(1)$                   | Constant      |
| $\Theta(\log(n))$             | Logarithm     |
| $\Theta(\log^c(n)), c \geq 1$ | Polylogarithm |
| $\Theta(n^a), 0 < a < 1$      | Sublinear     |

|                         |              |
|-------------------------|--------------|
| $\Theta(n)$             | Linear       |
| $\Theta(n \log(n))$     | Linearithmic |
| $\Theta(n^2)$           | Quadratic    |
| $\Theta(n^c), c \geq 1$ | Polynomial   |
| $\Theta(c^n), c > 1$    | Exponential  |
| $\Theta(n!)$            | Factorial    |



$\ln^2(n)$  grows slower than  $n^{0.5}$ . At around  $n = 56000$ ,  $\ln^2(n)$  is less than  $n^{0.5}$ .  
 $x^4$  grows slower than  $2^n$ . At around  $n = 16$ ,  $x^4$  is less than  $2^n$ .

### Beware

- It is **wrong** to say that `vector::insert` is  $\Theta(n)$ 
  - Because there is a case that it grows slower than  $n$
- It is **wrong** to say that `vector::push_back` is  $\Theta(n)$ 
  - Because there is a case that it grows slower than  $n$
- It is **ok** to say that `std::count` is  $\mathcal{O}(n)$ 
  - Because while it always grows as  $n$ , it **does not grow faster than  $n$**
  - $\mathcal{O}(n)$  is upper bound
  - But it is better to say that `std::count` is  $\Theta(n)$

### How to analyze using asymptotic notation

- Write a code
- Calculate the function  $f(n)$  that counts the number of instruction of the code when the data is size  $n$ 
  - Usually, just focus on **most executed line**
- Find  $g(n)$  and a notation  $X$  such that  $f(n)$  is  $X(g(n))$ 
  - It's either **Big-O** or **Big-Theta**. If there is a case that it can grow slower than  $G(n)$ , use Big-O

### Another Example

- Let's analyze `vector::push_back`

```
iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize + 1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1]; // <----- Most executed line: 1 (b)
    }
    mData[pos] = element;
    mSize++;
}
```

```

    return begin()+pos;
}
void push_back(const T& element) {
    insert(end(), element);
}

void expand(size_t capacity) {
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i]; // <----- Most executed line: 0, 0, 0, 0, 2, 4, 8, 16 (a)
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}

void ensureCapacity(size_t capacity) {
    if (capacity > mCap) {
        size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
        expand(s);
    }
}

// (a) can be 0 to n
// (b) can also be 0 to n
// Best case 0 + 0
// Worst case n + n
// vector::push_back is O(n)

```

### Another Definition for $\mathcal{O}$ and $\Theta$

- Using set builder notation
- $\mathcal{O}(g(n)) = \{f(n) \mid \text{there exists } c > 0 \text{ and } n_0 \geq 0 \text{ such that } f(n) \leq cg(n) \text{ for } n \geq n_0\}$
- $\Theta(g(n)) = \{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- The result is the same as definition using lim

### Summary

- We use Asymptotic Notation to describe efficiency of a program
  - Measure instruction count instead of time
  - Focus on growth rate of instruction count
- Find most frequently executed line in the code and count it
- Maps to Big-Theta if we have tight bound
- Use Big-O if we have upper bound

### Example

**Showing  $\log_2(n!)$  is  $\Theta(n \log_2(n))$**

- Will use set definition of Big-Theta
  - $\{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- Need to find  $c_1, c_2$  and  $n_0$

### Finding $c_1$ and $c_2$

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1$$

$$n! \leq n \times n \times n \times n \times \dots \times n$$

$$\log n! \leq \log n^n \quad \boxed{\log n^n = n \log n \text{ when } n \geq 1}$$

Found  $c_2 = 1$  for upper bound  $\log n! \leq n \log n$  when  $n \geq 1$

$$n! = n \times (n-1) \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times \left( \left\lfloor \frac{n}{2} \right\rfloor - 1 \right) \times \dots \times 1$$

$$n! \geq \left\lfloor \frac{n}{2} \right\rfloor \times \left\lfloor \frac{n}{2} \right\rfloor \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times 1 \times \dots \times 1$$

$$n! \geq \left( \frac{n}{2} \right)^{n/2} \quad \log \left( \frac{n}{2} \right)^{n/2} = \frac{n}{2} \log n - \frac{n}{2}$$

$$\log n! \geq \log \left( \frac{n}{2} \right)^{n/2} \quad 0.5n \log n - 0.5n \geq 0.4n \log n \text{ when } n \geq 32$$

Found  $c_1 = 0.4$  for lower bound  $\log n! \geq 0.4n \log n$  when  $n \geq 32$

$$0.4n \log n \leq \log n! \leq n \log n \text{ when } n \geq 32, c_1 = 0.4, c_2 = 1, n_0 = 32$$

$$\Rightarrow \log n! \text{ is } \Theta(n \log n) \text{ and } n \log n \text{ is } \Theta(\log n!)$$

## Lecture 13

### Priority Queue Simple Implementation

Featuring Binary Heap

#### Overview

- Simple implementation of `priority_queue`
- Quick intro to **Graph** and **Tree**
- **Binary Heap**
- `priority_queue` with **Binary Heap**

#### `priority_queue`

- Queue by value
- Max-in-First-Out

```
int main() {
    priority_queue<int> pq;
    pq.push(4);
    pq.push(20);
    pq.push(3);

    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

#### Vo.1, `priority_queue` by vector

- Use **vector** to store data
- push = simply `push_back`
- top, pop = find the max value and return / erase
- `max_element` returns iterator to max element

```
namespace CP {
    template <typename T>
    class priority_queue {
    protected:
        std::vector<T> v;
    public:
        bool empty(); { return v.empty(); }
        bool size(); { return v.size(); }
```

```
void push(const T &e) { v.push_back(e); }

T top() { return *std::max_element(v.begin(), v.end()); }

void pop() { v.erase(std::max_element(v.begin(), end())); }
};
}
```

#### **max\_element**

```
iterator max_element(iterator first, iterator last) {
    if (first == last) return last;
    iterator largest = first;
    ++first;
    for (; first != last; ++first)
        if (*largest < *first)
            largest = first;
    return largest;
}
```

#### **Vo.1 complexities**

|       |                             |
|-------|-----------------------------|
| push  | $\mathcal{O}(1)$ *amortized |
| top() | $\Theta(n)$                 |
| pop() | $\Theta(n)$                 |